

NEXUS — ENGINEERING INTELLIGENCE PLATFORM

Modernisation and Build Strategy v2.0

Version: 2.0

Status: Draft for Management Review

Prepared by: Engineering / IT Modernisation Programme — Scanfil APAC Manufacturing

Changes from v1.0:

- Corrected data authority model — Movex remains source of truth for committed BOMs; platform manages change workflow and draft state
- Corrected Stargile ERP integration — Stargile has its own Java MI interface, not movex-rest-api; Phase 1 Track A must produce an MI gap analysis
- Added two-site deployment architecture — site-aware ERP adapter configuration for Melbourne Site (Movex permanent) and JB Site (Movex now, IFS later); future country split handled by deployment configuration, not platform changes

Executive Summary

Scanfil APAC Manufacturing operates two independent legacy systems — Stargile (Java ECN) and PLMServer (PHP BOM and supplier intelligence) — to manage what is fundamentally a single, interconnected workflow. The NEXUS Engineering Intelligence Platform replaces both systems with a unified, AI-ready platform that adds a rich engineering change and intelligence layer on top of Movex, while preserving Movex as the source of truth for committed production BOMs.

Three key facts confirmed for v2.0:

First, **Stargile has its own Java MI interface** that calls the Movex MI API directly — it does not use `movex-rest-api`. Phase 1 Track A must map Stargile's Java MI call inventory and produce a gap analysis against `movex-rest-api`'s current coverage. Any missing MI endpoints must be added to `movex-rest-api` before Sprint 3 begins.

Second, **Movex remains the source of truth for committed production BOMs**. The platform manages the change workflow and draft state. When an ECN is approved, the platform pushes the committed revision to Movex via the ERP Adapter. Movex updates its production BOM record and

remains authoritative. The platform holds the rich history of every change, every approval, and every supplier evaluation — a layer of intelligence that Movex cannot provide — but it does not replace Movex as the system of record for what is in production.

Third, **Scanfil APAC operates two sites under the same entity**, both currently in one country. Melbourne Site will keep Movex permanently. JB Site uses Movex now but will migrate to IFS, and will later split across two countries. This is handled by **site-aware configuration** — not multi-tenancy. The same platform codebase runs at both sites; environment variables control which ERP adapter each site uses. When JB Site splits across two countries, that is a deployment configuration change, not an architectural one.

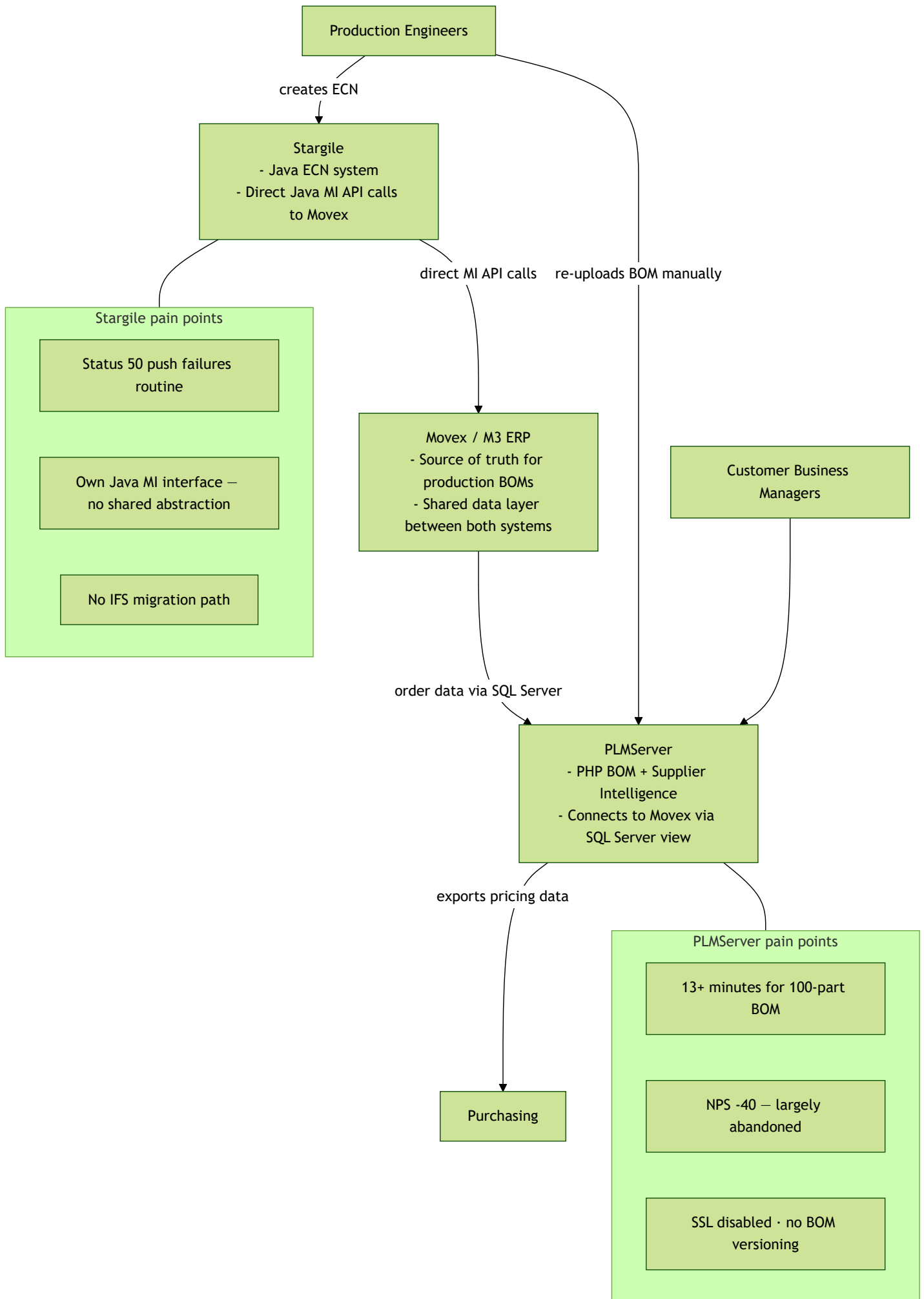
Iteration 1 — three modules, built together:

- **ECN Module** — replaces Stargile
- **BOM Module** — replaces PLMServer BOM management
- **Supplier Intelligence Module** — replaces PLMServer APIManager

Future iterations: Route Management → MSP → MES

2. Current Situation — Two Systems, One Problem

2.1 Current Architecture



The critical gap: there is no direct integration between Stargile and PLMServer. Engineers manually re-upload BOM data from Movex into PLMServer after completing an ECN. Two systems, two re-entry points, no shared event stream.

2.2 What Stargile's ERP Interface Actually Is

Stargile connects to Movex through its own Java classes that call the Movex MI API (M3 Business Engine API) directly. This is a Java-native MI integration, not the `movex-rest-api` service. This has two implications for the programme:

Phase 1 Track A implication: When mapping Stargile's ERP integration, the team must document each MI program called by Stargile's Java classes — the program name, the transaction type, the input fields used, the output fields consumed, and any known quirks. This MI call inventory is then cross-referenced against what `movex-rest-api` already implements. The delta becomes a list of new endpoints that must be added to `movex-rest-api` before the ECN module's Sprint 3 ERP integration work begins.

Architecture implication: The new ECN module will call `movex-rest-api` over HTTP — not MI directly. This is the correct design. It keeps one ERP boundary for the entire platform. But it means `movex-rest-api` must be extended to cover all ECN-relevant MI operations before the ECN module can be fully functional. This extension work is a Phase 1 → Phase 2 dependency that must be tracked explicitly.

2.3 The Engineer Workflow Today

Step	System	Pain
Create ECN, assign approvers	Stargile	Manual Excel upload. Status 50 push failures are routine — date conflicts require workaround.
Approval routing	Stargile	Manual — no automated reminders. Password confirmation per approver.
Commit BOM to Movex	Stargile (Java MI → Movex)	Manual revision update in Movex PDS screen after Status 60.
Re-upload BOM to PLMServer	PLMServer	Manual re-entry or file upload — no automatic sync.
Run supplier pricing	PLMServer	13+ minutes for 100-part BOM. No progress feedback.
Review and export	PLMServer	Manual Excel export. No BOM comparison. NPS -40.

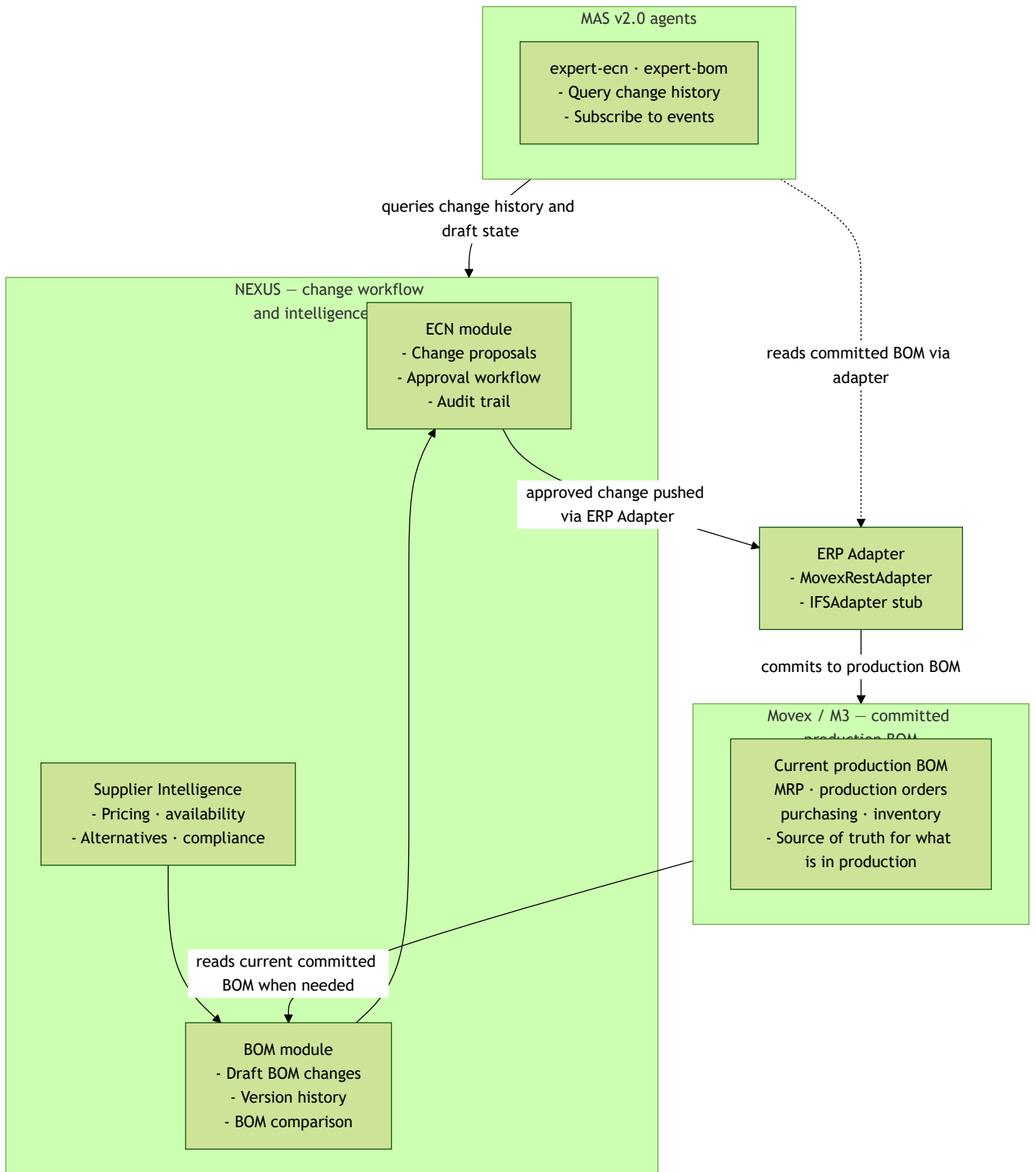
3. Data Authority Model — The Correct Architecture

3.1 The Key Distinction

The v1.0 strategy used the phrase "the platform owns the BOM." This was imprecise and warrants correction before it shapes design decisions.

In a manufacturing operation — particularly one under ISO 13485 — Movex drives MRP, production orders, purchasing, and inventory. Other departments update BOMs in Movex outside the ECN process. If the platform claimed to own the BOM and Movex was downstream, the platform would immediately face a bidirectional synchronisation problem: Movex changes something, the platform does not know, and the platform's copy becomes stale.

The correct model is **layered authority**:



3.2 What Each Layer Owns

Data	Authority	Lives in	Why
Committed production BOM — what is currently in production	Movex	Movex	MRP, production orders, purchasing all depend on Movex being authoritative. Auditors expect it.
Proposed BOM change — the ECN draft	Nexus	Nexus database	Change workflow and approval management is the Nexus's core function.
ECN history — every change ever raised	Nexus	Nexus database	ISO 13485 audit trail. Immutable. Agents query this.
BOM version chain — history of revisions	Nexus	Nexus database	Rich history the platform accumulates from each ECN commit. Not available in Movex.
Supplier intelligence — pricing, availability	Nexus	Nexus database (Redis cache + PostgreSQL)	Movex has no supplier API integration.
Compliance flags — RoHS/REACH/WEEE	Nexus	Nexus database	Populated from Octopart and manual entry.

3.3 Arguments for This Model

No synchronisation risk. Movex can be updated by other processes — purchasing corrections, direct BOM edits by other users — without breaking the platform's integrity. The platform records what it pushed; it does not claim to know the current Movex state beyond its own commits.

ISO 13485 aligned. The validated system of record for the medical device BOM is Movex. ISO 13485 auditors expect Movex to be authoritative for the production BOM. The platform provides the change traceability layer on top of it — which is exactly what the standard requires.

Multi-site safe. Both Melbourne Site and JB Site keep Movex as their production authority. The ERP Adapter controls where each site commits. When JB Site migrates to IFS, its production authority changes to IFS, but the model is unchanged — the platform still pushes approved changes through the adapter; it simply pushes to a different target.

AI platform value is unchanged. The change history, draft proposals, supplier intelligence, approval records, and BOM version chain all live in the platform. MAS v2.0 agents query the platform for engineering intelligence. They read the committed BOM from Movex via the adapter only when the current production state is needed — which is a minority of agent queries.

4. Two-Site Deployment Architecture

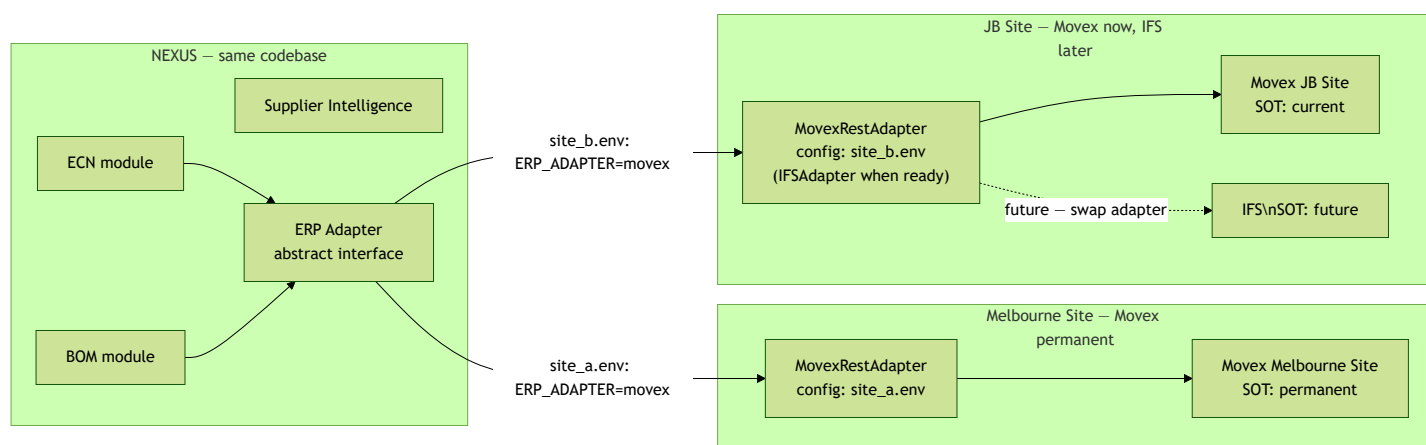
4.1 The Requirement

Scanfil APAC currently operates two sites, both in the same country, both under the same Scanfil APAC entity beneath the Scandinavian parent. Their ERP futures differ:

- **Melbourne Site:** Movex permanently. No planned IFS migration.
- **JB Site:** Movex now, IFS migration planned. JB Site will also split across two countries in future.

4.2 Why This Is a Configuration Problem, Not an Architecture Problem

Because both sites are the same Scanfil APAC entity, the platform does not need multi-tenancy — separate databases, separate schemas, separate authentication domains. It needs **site-aware ERP adapter configuration**. The same platform codebase determines which ERP adapter to use based on environment variables set at deployment time.



4.3 Deployment Options

The two sites can run as:

Option 1 — Two instances, shared codebase (recommended): Each site runs its own Docker Compose stack pointing to its own PostgreSQL instance and its own ERP adapter configuration. Data is separated at the instance level. The codebase is identical — deployed from the same Git

repository, same Docker images, different environment files. This is the cleanest separation for ISO 13485 purposes — each site has an independent Software Validation record.

Option 2 — Single instance, shared database with site partition: Both sites share one platform deployment. A `site_id` column partitions data. A single Software Validation record covers both sites. More operationally efficient but carries higher risk — a deployment issue affects both sites simultaneously.

Option 1 is recommended. The operational overhead of two instances is low given Docker's deployment repeatability, and the ISO 13485 independence is valuable.

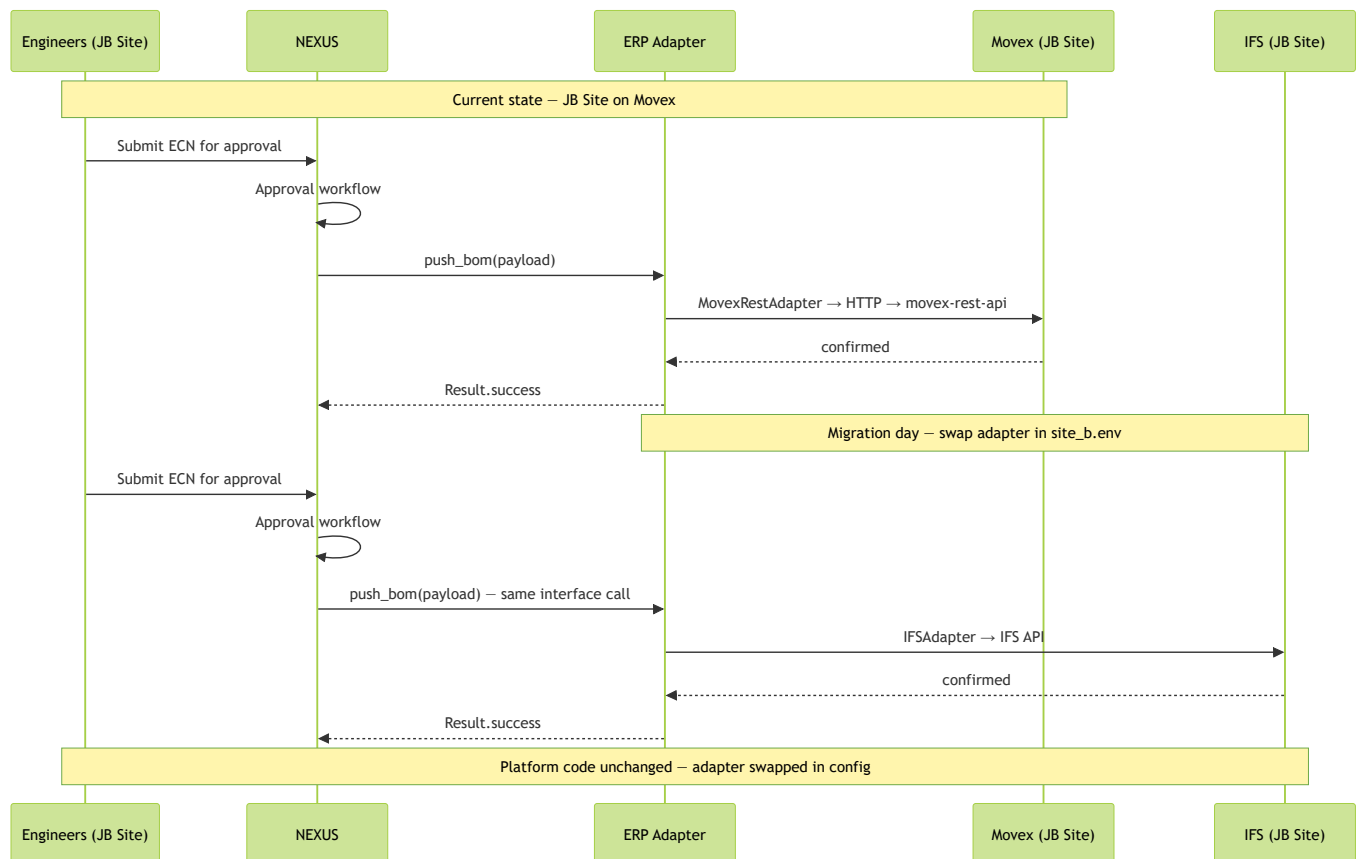
4.4 When JB Site Splits Across Two Countries

When JB Site eventually becomes two separate country operations, the response is:

1. Clone the JB Site Docker deployment into two instances — JB Site-Country 1 and JB Site-Country 2
2. Each gets its own environment file with its own ERP target configuration
3. Historical data migration: a point-in-time copy of JB Site's database is made; each new instance starts from that copy; historical records are read-only
4. No platform code changes required

The platform architecture accommodates this entirely through configuration and deployment. The IFS adapter stub built in Sprint 4 validates that the interface is complete for both Movex and IFS operations — including any differences in how IFS handles the MI operations that were previously done via Stargile's Java interface.

4.5 JB Site IFS Migration Sequence



5. Why One Unified Platform — Not Two Separate Replacements

5.1 The Data Model Argument

The BOM is the central entity in both Stargile and PLMServer. An ECN proposes a change to a BOM. PLMServer evaluates a BOM against supplier availability. Both need to work on the same draft BOM record. Today they cannot — they communicate only through the committed Movex BOM, which means engineers must manually re-upload after completing an ECN.

In the platform, an ECN works on a draft BOM record in the same database. The supplier intelligence module reads the same draft BOM. When the ECN is approved, the adapter pushes the committed revision to Movex. This design only works if ECN, BOM, and Supplier Intelligence share one data model.

5.2 The Supplier Intelligence Argument

When an engineer is creating an ECN to add a new component, the most useful thing the system can do is show — immediately, in the same screen — whether that component is available, at what price

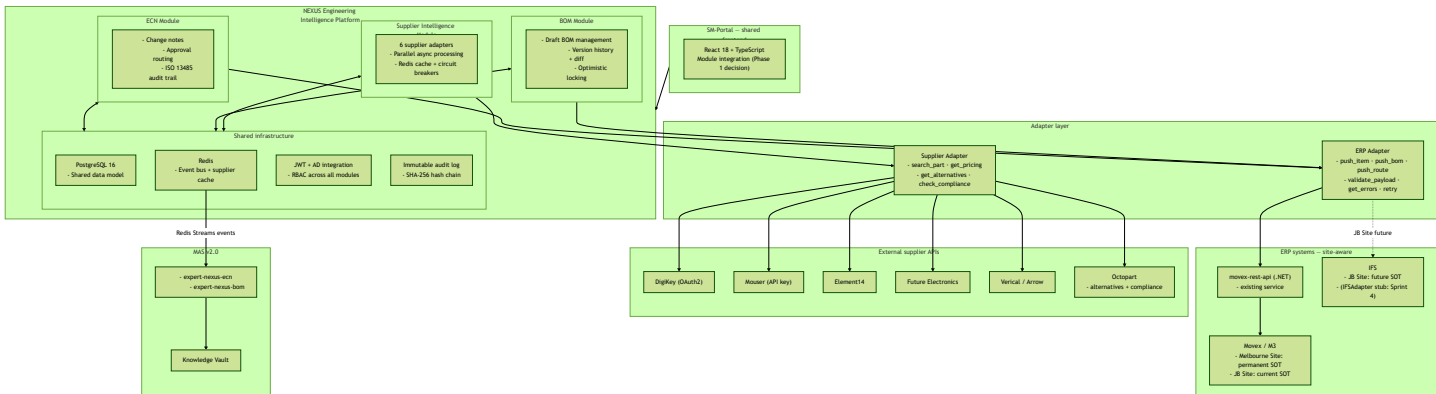
and lead time, and whether an alternative exists if it is not. This is only possible if the ECN editor and the Supplier Intelligence module share the same session, the same draft BOM, and the same UI.

5.3 The IFS Migration Argument

Building ECN and BOM as separate replacements would mean each system needs its own ERP integration. When JB Site migrates to IFS, both systems must be updated. Building them as one platform with one ERP Adapter means the migration work happens once — in one adapter implementation.

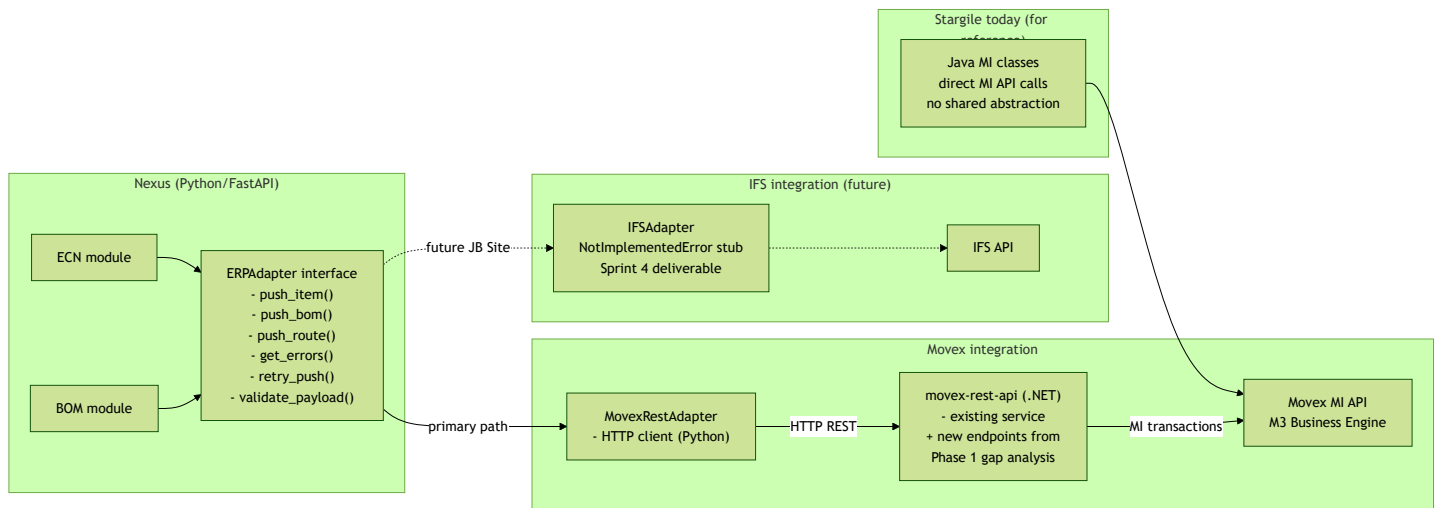
6. Platform Architecture

6.1 Architecture Overview



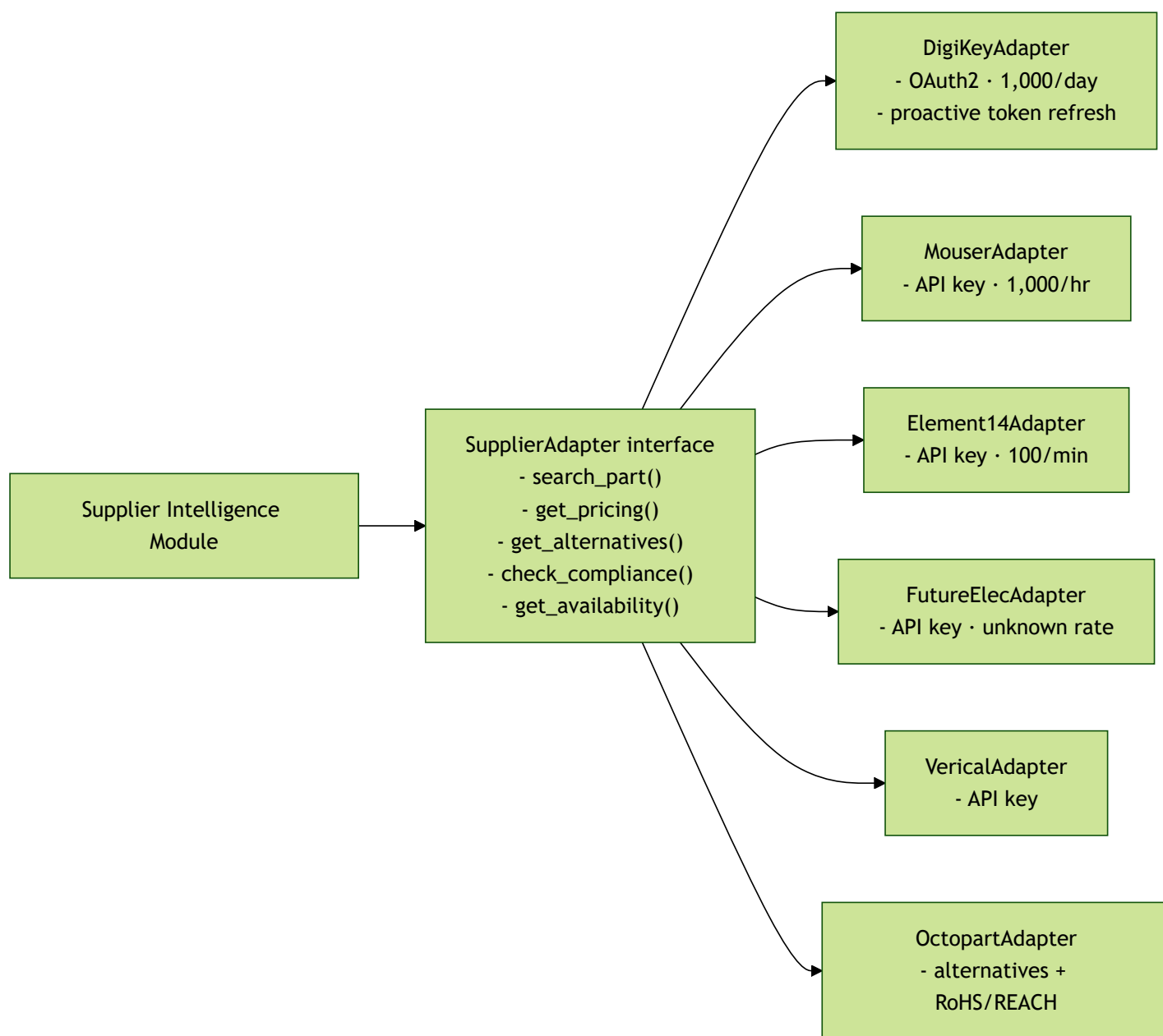
6.2 The ERP Adapter — Corrected for Stargile's MI Interface

The ERP Adapter sits between the platform's ECN and BOM modules and any ERP system. In v2.0, the implementation path is clarified:



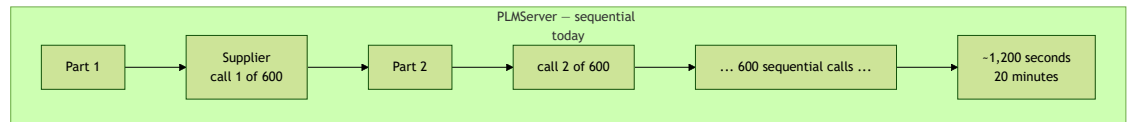
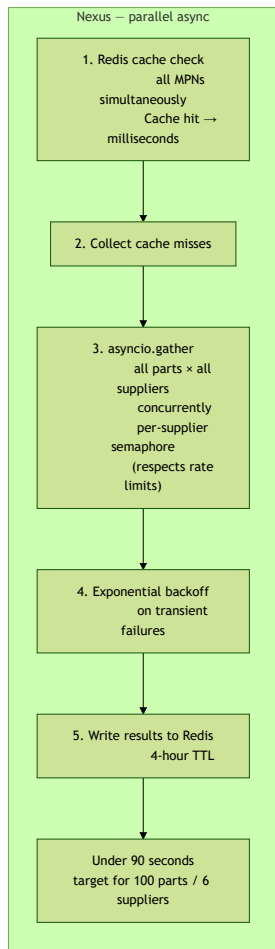
The critical Phase 1 Track A deliverable: Stargile's Java MI classes must be analysed to extract the complete MI program call inventory — every MI program, transaction type, input fields, and output fields used. This inventory is cross-referenced against `movex-rest-api`'s existing endpoints. The gap — MI operations Stargile uses that `movex-rest-api` does not currently expose — becomes a list of new endpoints that the `.NET` team must add to `movex-rest-api` before Sprint 3. This gap analysis is a Phase 1 gate deliverable, and the endpoint additions are a Phase 2-to-Sprint 3 dependency.

6.3 The Supplier Adapter



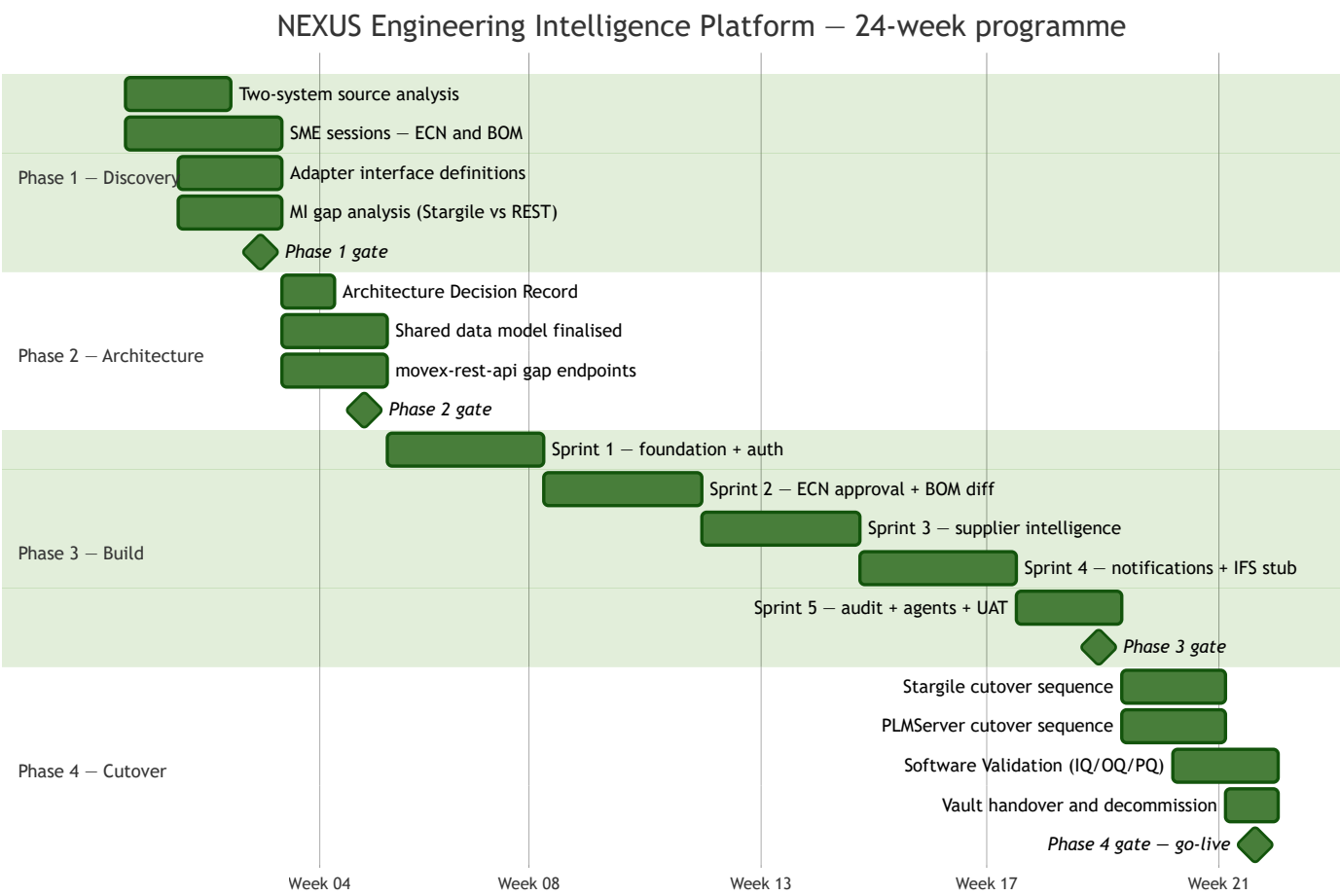
Each adapter is independently circuit-breakered. If DigiKey is unavailable, the remaining five adapters continue. Stale cached results are returned with a staleness warning rather than an error. OAuth tokens for DigiKey are refreshed proactively by the Celery worker — eliminating PLMServer's crash-on-token-expiry failure mode.

6.4 Async Parallel Processing — The Core Performance Fix



Redis serves dual purpose: event bus (Redis Streams for MAS v2.0) and supplier cache (Redis key-value with 4-hour TTL). A Celery worker handles the long-running supplier API fan-out in the background — publishing real-time progress per part via WebSocket to the engineer's browser. This directly solves PLMServer's "system appears frozen" problem.

7. Four-Phase Programme — 24 Weeks



7.1 Phase 1 — Discovery (Weeks 1–4)

Phase 1 covers two legacy systems and three adapter interfaces. The `movex-rest-api` gap analysis is a new Phase 1 deliverable that did not exist in v1.0.

Track A — Source Code and System Analysis

For PLMServer (source available): the January 2026 analysis at `analysis/PLMServer_Complete_Analysis.md` and `analysis/plm-tables.sql` are the primary inputs. Validate the analysis against live system behaviour. Confirm all six supplier integrations, OAuth token lifecycle, rate limits, and price break algorithms.

For Stargile (source pending — permission being sought for extraction from server): once source access is confirmed, execute the following specific analysis:

1. Extract the complete list of MI programs called by Stargile's Java classes — program name, transaction type, all input and output fields used, any hardcoded values or quirks
2. Cross-reference this inventory against `movex-rest-api` 's existing endpoint coverage
3. Document the gap: MI operations Stargile uses that `movex-rest-api` does not expose

4. Produce a specification of new endpoints required in `movex-rest-api` — this spec goes to the `.NET` team as a Phase 2 dependency

If Stargile source access is not confirmed by end of Week 2, the gap analysis proceeds via SME sessions (Track B) and direct MI API documentation — the source accelerates this but does not gate it.

Before any source file is opened, run `/m3-lookup` in the Knowledge Vault. The vault's `m3-knowledge/transactions/` directory documents MI programs from real production experience. This is the starting point.

Track B — SME Behavioural Sessions

Stargile sessions (Production Engineers, Document Control, SMT Engineers): complete ECN state machine — specifically resolving the Status 60 exit mechanism; all exception paths; Excel template decision; SMT programme workflow.

PLMServer sessions (Engineers, CBMs, Purchasing): confirm which workflows are genuinely used (20% adoption — understand why the others stopped); full supplier list and regional requirements; RoHS/REACH/WEEE reporting requirements; BOM approval and export workflow.

Shared sessions: data retention decisions for both systems; SM-Portal integration architecture review; two-site deployment confirmation with IT.

Track C — Three Adapter Interface Definitions

- ERP Adapter: `push_item` , `push_bom` , `push_route` , `get_errors` , `retry_push` , `validate_payload` — calls `movex-rest-api` over HTTP
- Supplier Adapter: `search_part` , `get_pricing` , `get_alternatives` , `check_compliance` , `get_availability` — one implementation per supplier
- BOM Comparison Service: `compare(bom_a_id, bom_b_id) → DiffResult`

Phase 1 Gate Deliverables

Deliverable	Owner	Sign-off
ECN Behavioural Specification (full state machine, all roles, exception paths)	Developer + SMEs	Production Engineer, Document Control
BOM/Supplier Behavioural Specification	Developer + SMEs	Engineer, CBM, Purchasing
ERP Adapter Interface	Developer	IT / Architecture
Supplier Adapter Interface (all six suppliers)	Developer	IT / Architecture

Deliverable	Owner	Sign-off
BOM Comparison Service Interface	Developer	IT / Architecture
Stargile MI gap analysis + movex-rest-api extension spec	Developer	IT / .NET team
Retired Functionality Lists (one per legacy system)	Developer + SMEs	Manager + SMEs
Excel template decision	SMEs	Production Engineer
SM-Portal integration decision	Developer + IT	IT, Manager
Data retention decisions (both systems)	Developer + QA	QA sign-off
Two-site deployment model confirmed	Developer + IT	IT, Manager
Vault entries (both projects, compliance notes, ADRs)	Developer	Vault commit

7.2 Phase 2 — Architecture Decision and Build Plan (Weeks 5–6)

Phase 2 gate conditions — all must be met before Phase 3 begins:

Gate Condition	Owner	If Not Met
Architecture Decision Record complete	Developer	Extend Phase 2
Shared data model finalised	Developer	Extend Phase 2
Non-Functional Requirements baselined	Developer + IT	Extend Phase 2
movex-rest-api gap endpoints specified and work scheduled	.NET team + Developer	Sprint 3 ERP integration cannot begin
Movex test/sandbox environment confirmed	IT	Phase 3 cannot begin
Docker on Windows Server confirmed	IT	Phase 3 cannot begin
Supplier API test credentials confirmed (all 6)	IT	Sprint 3 supplier work cannot begin

Gate Condition	Owner	If Not Met
Two-site deployment model agreed	IT + Manager	Sprint 1 environment setup cannot be finalised
All data retention decisions signed	QA	Phase 3 cannot begin

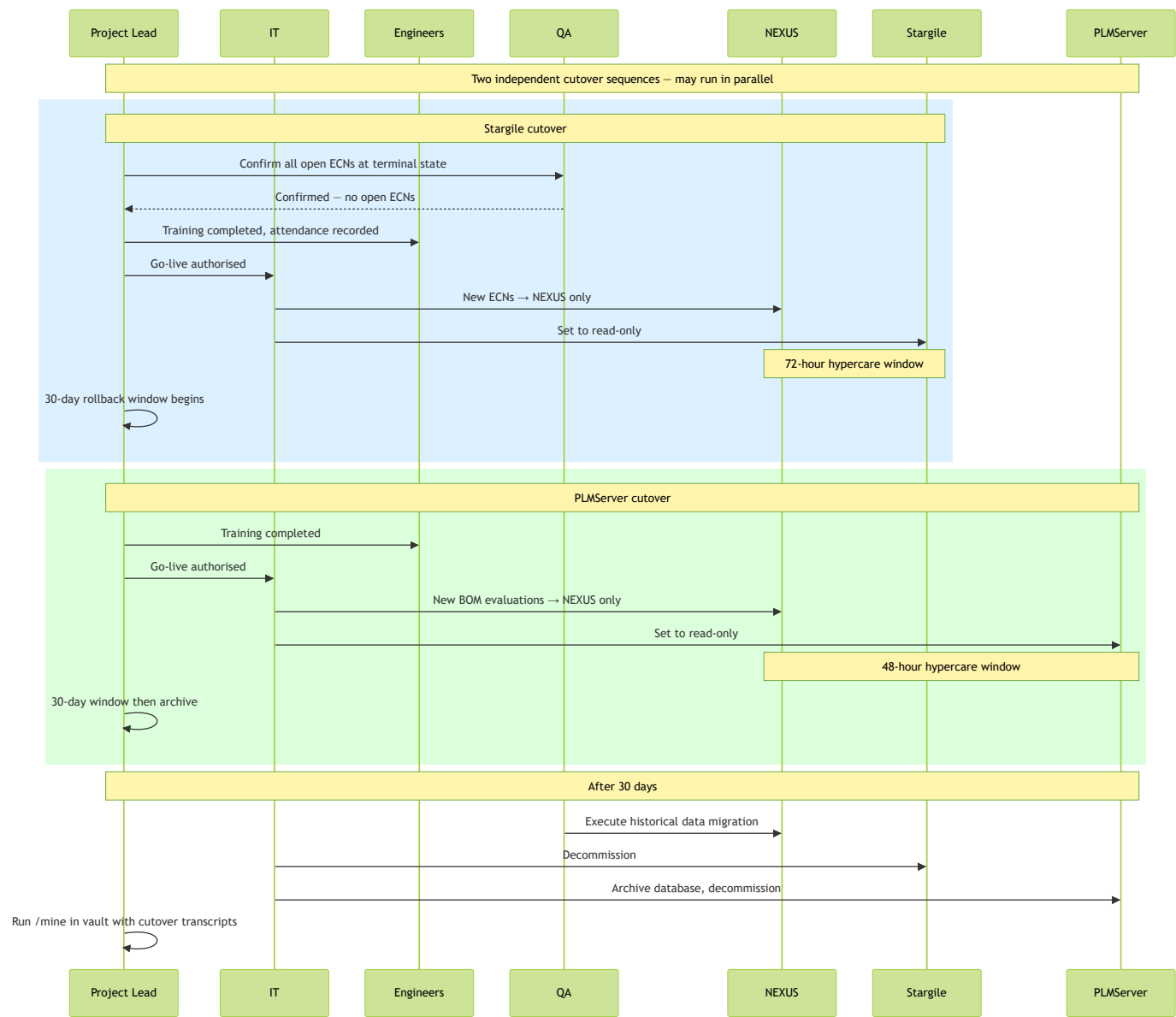
7.3 Phase 3 — Build (Weeks 7–20)



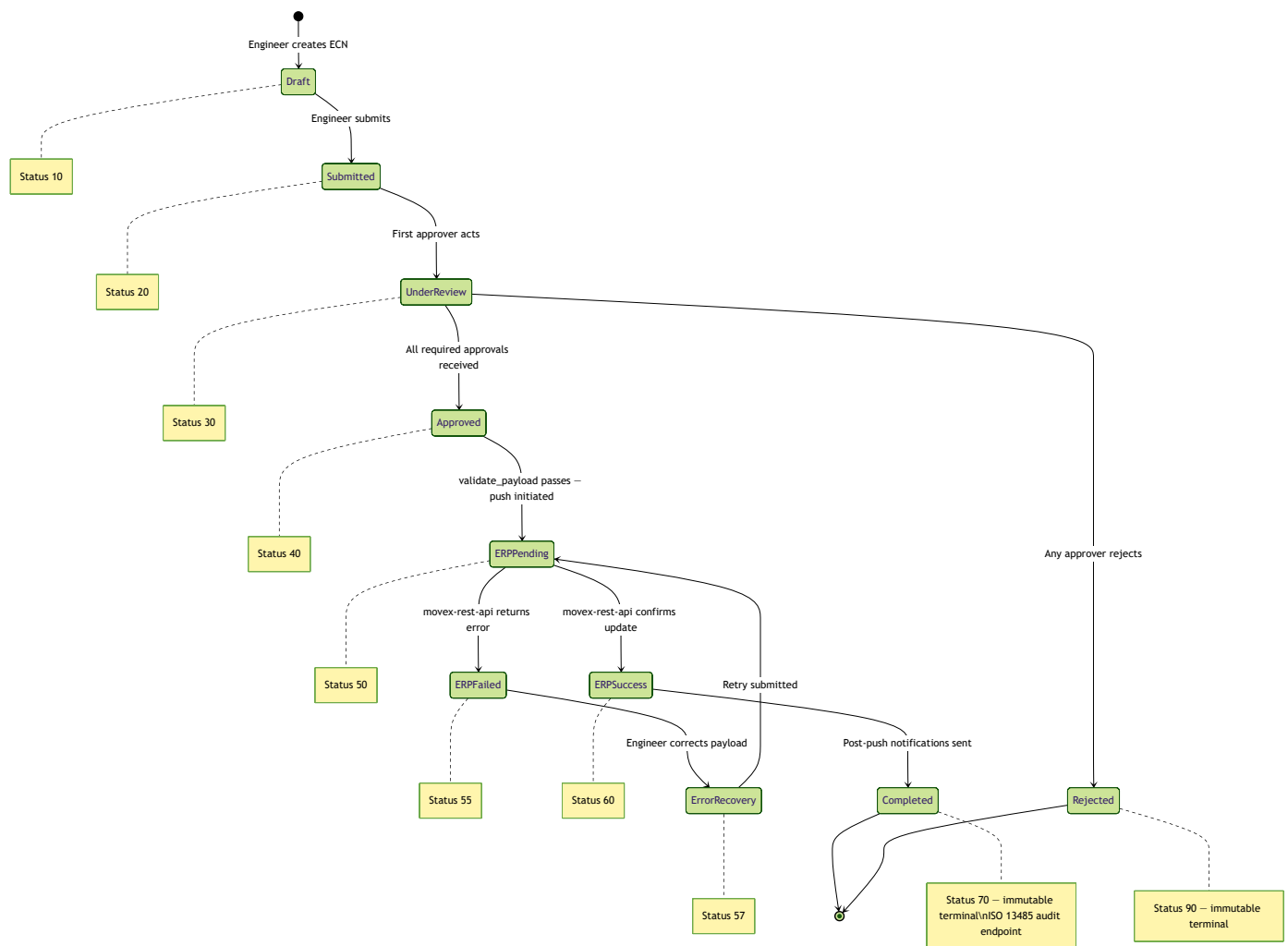
Sprint 3 mandatory milestone — performance demonstration: Before Sprint 5 UAT, demonstrate 100-part BOM processing to engineers with real data. PLMServer's NPS is -40 because engineers were burned by 13-minute processing. Show the sub-90-second result before asking engineers to trust the new system. This is a Sprint 3 definition of done item, not an optional communication activity.

Sprint 4 deliverable — IFS Adapter Stub: Implements all `ERPAdapter` interface methods with `NotImplementedError` and a descriptive message. Deployed behind a feature flag, disabled in production. Purpose: validate the interface contract is complete and sufficient for IFS operations before any IFS integration work begins. Any interface gaps found during stub design are fixed in Sprint 4. JB Site environment configuration (pointing to the stub for testing) is also a Sprint 4 deliverable.

7.4 Phase 4 — Cutover (Weeks 21–24)



8. ECN State Machine



Status 70 — Completed is a new explicit terminal state confirmed in v2.0. The April 2018 transcript showed Status 60 as the apparent endpoint, but did not confirm whether a further terminal state existed. Phase 1 Track A must resolve this from Stargile's Java source. If Stargile already has an equivalent terminal state, Status 70 maps to it. If Status 60 was de facto terminal, the state machine collapses Status 60 and 70 into one state with the immutability semantics of Status 70.

validate_payload is called before the ERP push is initiated — before the ECN reaches Status 50. It pre-checks effective date conflicts (the known date-conflict class of Status 50 failures) and any payload validation the `movex-rest-api` endpoint enforces. If validation fails, the engineer receives a clear, actionable error message before any MI transaction is attempted.

9. Non-Functional Requirements

9.1 Performance

Requirement	Target	Source
Page load time	< 3 seconds on local network	Workstation standard
API response (standard operations)	< 500ms at 95th percentile	Workflow productivity
API response (ERP push)	< 30 seconds	movex-rest-api SLA
BOM processing — 100 parts / 6 suppliers	< 90 seconds	vs 13+ minutes today
BOM processing — 50 parts / 6 suppliers	< 45 seconds	Typical customer order
Supplier cache hit rate	> 70% for repeat MPNs	Redis 4-hour TTL
BOM comparison (500-line BOM)	< 3 seconds	Diff algorithm target
Audit log query (12 months)	< 10 seconds	ISO 13485 review
Concurrent users	20 simultaneous without degradation	7–8 engineers + CBMs + Purchasing

9.2 Availability and Reliability

Requirement	Target
Uptime (production)	99% Mon–Fri 07:00–18:00
Planned downtime	Off-shift — weekends and overnight
RTO	4 hours
RPO	Maximum 1 hour
Supplier degraded mode	System usable if 1–2 suppliers unavailable — per-supplier circuit breaker

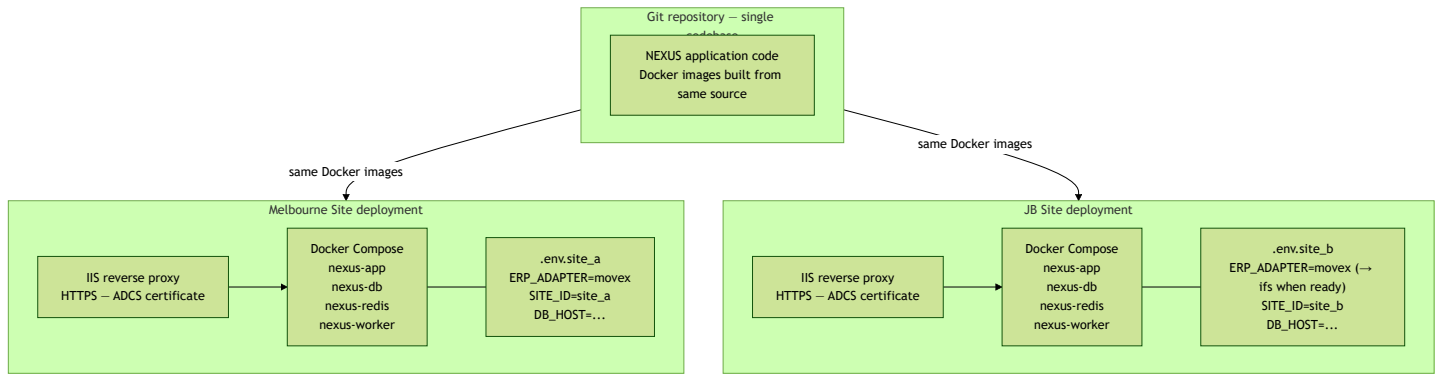
9.3 Scalability

Requirement	Target
User growth	100 users without architectural change — accommodates group rollout
BOM record volume	50,000 BOMs with full version history
Part (MPN) catalogue	500,000 parts without query degradation
Site instances	Designed for 2 instances; extendable to additional Scanfil APAC sites by configuration

10. Infrastructure and Hosting

10.1 Deployment Model

On-premise, Windows Server. Docker on Windows Server via WSL2. Consistent with `movex-rest-api` , `SM-Portal`, and `MyInvois-Service`. Two instances — one per site — deployed from the same Docker images with different environment files.



10.2 Docker Compose Services

Service	Image	Purpose
nexus-app	Python/FastAPI (custom)	All three Iteration 1 modules via FastAPI routers
nexus-db	PostgreSQL 16	Shared data model per site instance
nexus-redis	Redis 7	Event bus (Redis Streams) + supplier cache (key-value)

Service	Image	Purpose
nexus-worker	Python Celery + Redis	Background supplier API tasks, OAuth token refresh

10.3 Server Architecture

Server	Role	Minimum Spec
App Server (per site)	Docker host — app, redis, worker; IIS reverse proxy on host	Windows Server 2019+, 16 cores, 32 GB RAM, 200 GB SSD
DB Server (per site)	PostgreSQL 16	Windows Server 2019+, 8 cores, 32 GB RAM, 1 TB SSD

10.4 Backup and Recovery

Backup Type	Schedule	Retention
PostgreSQL full backup	Nightly at 01:00	30 days
PostgreSQL WAL archive	Continuous / hourly	7 days
Redis snapshot (RDB)	Every 4 hours	7 days (rebuildable)
Docker images	On every deployment	Last 5 per service
Application code	Continuous	Full Git history + vault commit hooks

11. Security Design

11.1 PLMServer Security Issues — All Corrected by Design

PLMServer Issue	Severity	Resolution
HTTPS disabled	Critical	IIS enforces HTTPS with ADCS certificate
SSL verification disabled (<code>verify: false</code>)	Critical	All Python HTTP clients use <code>verify=True</code>
Raw SQL queries — injection risk	High	SQLAlchemy ORM parameterised statements

PLMServer Issue	Severity	Resolution
OAuth tokens not refreshed proactively	High	Celery worker refreshes DigiKey tokens 5 min before expiry
No rate limiting	Medium	Per-supplier semaphore with configurable limits
No input validation	High	Pydantic models validate all inputs server-side

11.2 Authentication

JWT tokens in HTTP-only cookies. Access tokens: 15 minutes. Refresh tokens: 8 hours with server-side revocation. Active Directory integration recommended — engineers use Windows credentials; leavers automatically locked out.

Approval actions require password re-confirmation at the moment of approval — ISO 13485 non-repudiation.

11.3 Role-Based Access Control

Permission	Production Engineer	Document Control	CBM	Purchasing	SMT/Test	Admin
Create / edit ECN	✓	—	—	—	—	—
Approve ECN	—	✓	—	—	✓	—
Create / edit BOM	✓	—	—	—	—	—
Upload BOM	✓	—	✓	—	—	—
Run supplier pricing	✓	—	✓	✓	—	—
Approve BOM for Purchasing	—	—	✓	—	—	—
Export BOM for ordering	—	—	—	✓	—	—
View compliance reports	—	✓	—	✓	—	✓

Permission	Production Engineer	Document Control	CBM	Purchasing	SMT/Test	Admin
View audit log	—	✓	—	—	—	✓
Manage users / roles	—	—	—	—	—	✓
Configure supplier APIs	—	—	—	—	—	✓

11.4 Audit Trail Security

-- Application user has INSERT + SELECT on audit_log only
REVOKE UPDATE, DELETE ON audit_log FROM nexus_app_user;

SHA-256 hash chain across all rows — any post-write modification breaks the chain. Validation query runs nightly as part of backup verification.

12. MAS v2.0 and Knowledge Vault Integration

12.1 Agent Registrations — Sprint 5 Deliverables

Two agents registered in c:/Projects/.github/agents/manifest.json before Phase 3 ends:

expert-nexus-ecn

Capability	Endpoint
Query ECN status by ID or customer ref	GET /ecns/{id}
Query approval state	GET /ecns/{id}/approvals
Subscribe to ECN events	Redis Streams ecn.*
Pre-validate a BOM change	POST /validate/bom

expert-nexus-bom

Capability	Endpoint
Query BOM draft and version history	GET /boms/{id}/history

Capability	Endpoint
Compare two BOM revisions	GET /boms/compare?a={id}&b={id}
Query supplier availability	GET /parts/{mpn}/availability
Query compliance flags	GET /parts/{mpn}/compliance
Subscribe to BOM events	Redis Streams bom.*

12.2 Knowledge Vault Integration Points

Phase	Action	Vault Skill
Phase 1 — before Track A	/m3-lookup for all known Stargile MI programs	/m3-lookup
Phase 1 — Track A output	Add PLMServer + Stargile project entries to vault/projects/	Manual
Phase 1 — Track A output	Add ECN (ISO 13485) + BOM (RoHS/REACH) compliance notes	Manual
Phase 1 — decisions	Record all adapter interface decisions as ADRs	/adr
Phase 1 — MI gap analysis	Record Stargile MI inventory and gap as ADR	/adr
Phase 2	Record full Architecture Decision Record	/adr
Phase 3 — Sprint 1	Install commit hook at c:/Projects/Nexus/	install-hooks.sh
Phase 3 — Sprint 5	Register both agents in agent manifest	Manual
Phase 4	/mine both cutover transcripts; /runbook rollback procedures	/mine /runbook
Phase 4 — month 1	/review to capture early operational patterns	/review

13. Compliance

13.1 Two Compliance Regimes

Regime	Scope	Primary Module	Key Requirements
ISO 13485	Medical device manufacturing	ECN Module	Immutable audit trail; non-repudiable approvals; IQ/OQ/PQ Software Validation; record retention for device lifetime
RoHS / REACH / WEEE	EU and APAC environmental	BOM + Supplier	Compliance flags per component; enforced at BOM edit; reportable per BOM line
ISO 9001	General quality management	Platform-wide	Audit trail; document control; corrective action

13.2 Software Validation Documentation (ISO 13485 — ECN Module)

Document	Content	Evidence Source
IQ	System installed and configured correctly	Deployment runbook; Docker image digest
OQ	System performs as specified	ECN Behavioural Specification + ECN UAT Report
PQ	System performs correctly in production	Pilot period monitoring data

14. Change Management

14.1 Risk by System

System	Change Risk	Primary Driver	Key Action
Stargile replacement	LOW	Management mandate + user frustration	Track B involvement builds ownership
PLMServer replacement	MEDIUM	System abandoned — engineers gave up	Sprint 3 performance demo before UAT. Show 90-second processing with real data.

14.2 Stakeholder Map

Stakeholder	Role	Primary Interest	Key Engagement
Manager / Sponsor	Decision owner	Delivery, ROI, engineer retention	Phase gate reviews, bi-weekly updates
Production Engineers (7–8)	ECN + BOM primary users	Faster, less manual, less friction	Track B sessions, Sprint 3 demo, UAT
CBMs	BOM approvers	Clear approval interface	PLMServer Track B, UAT
Purchasing	BOM consumers	Accurate supplier export	Track B, UAT
Document Control / QA	Compliance owners	ISO 13485, RoHS/REACH audit trail	SME sessions, Software Validation
IT	Infrastructure	Docker-deployable, two-site supportable	Phase 2 architecture, provisioning

15. Risk Register

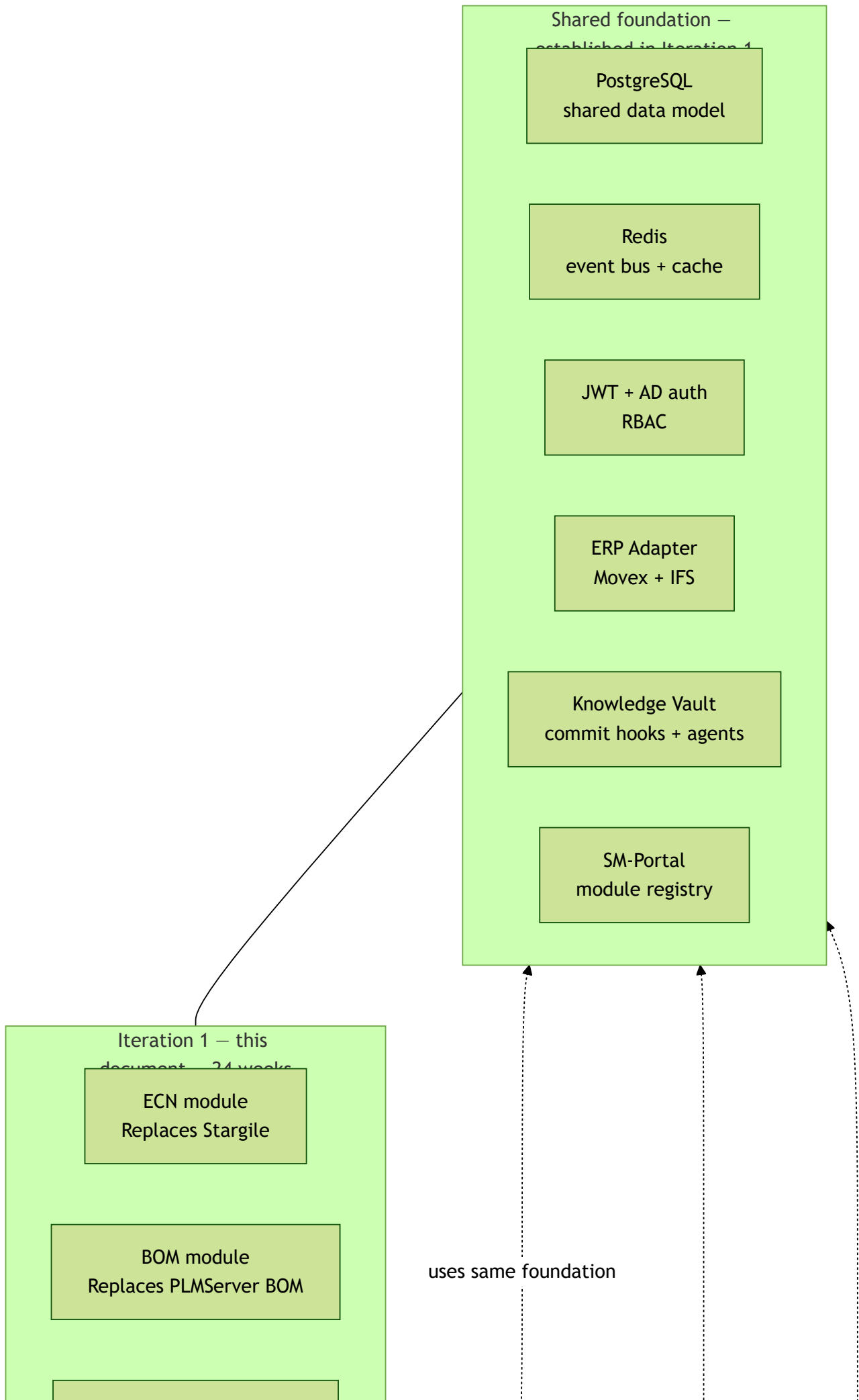
Risk	Impact	Likelihood	Mitigation
Stargile MI gap analysis reveals large extension	High	Possible	Phase 1 Track A produces the spec. .NET team scopes extension in Phase 2. If scope is large, Sprint 3 ERP integration is

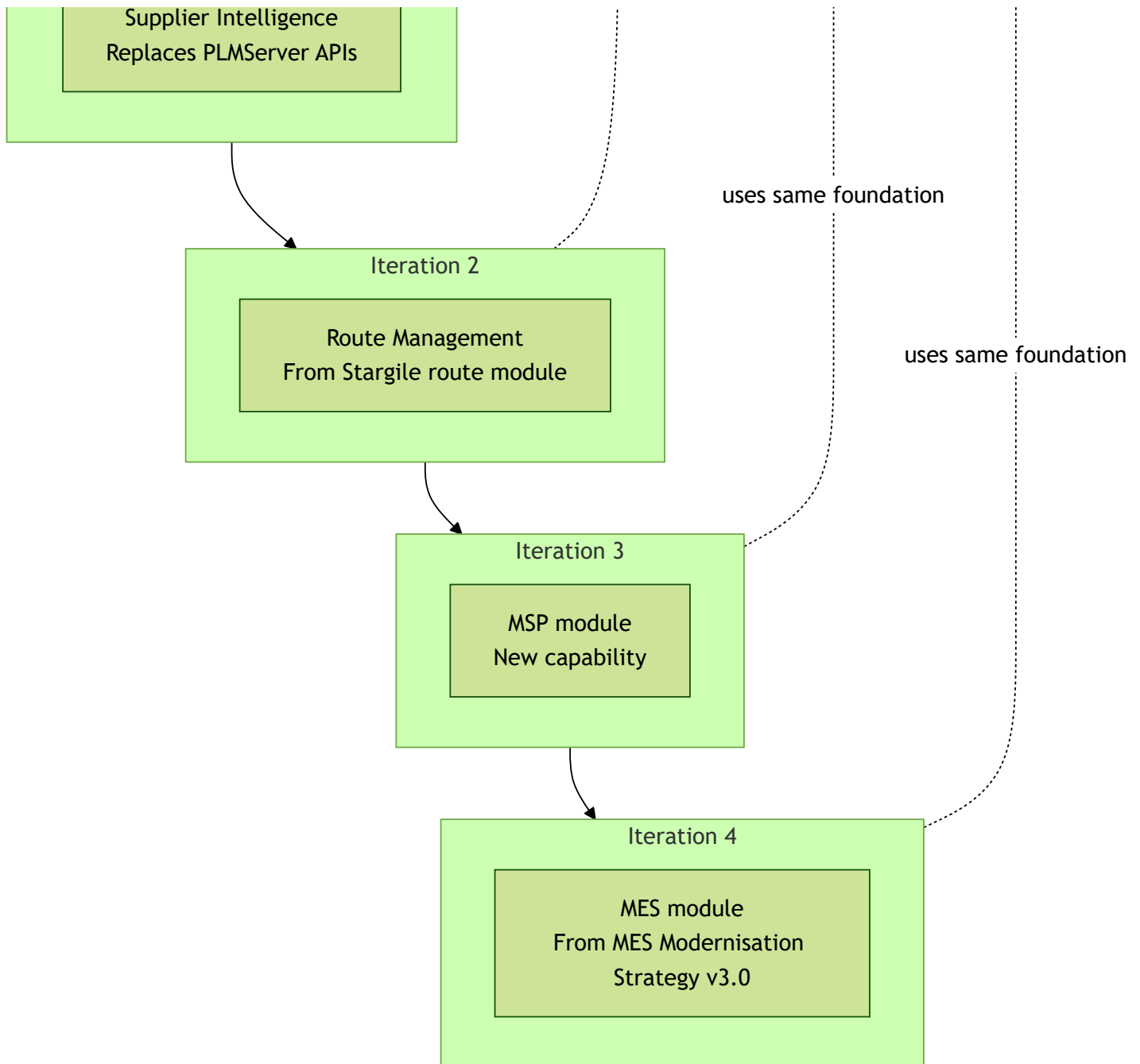
Risk	Impact	Likelihood	Mitigation
scope for movex-rest-api			sequenced after extensions are deployed to staging.
Stargile source code not obtained before Phase 2	High	Possible	SME sessions + MI API documentation cover Behavioural Spec without source. MI gap analysis proceeds via SME and direct MI documentation.
PLMServer engineer adoption low despite improvements	Medium	Medium	Sprint 3 performance demo before UAT. Show 90-second BOM processing with real data.
Supplier API test credentials not confirmed before Sprint 3	High	Possible	Phase 2 hard gate for all six suppliers. Sprint 3 cannot begin without them.
DigiKey OAuth token lifecycle issues persist	Medium	Possible	Celery worker proactive refresh. Tested against DigiKey sandbox in Sprint 3. PLMServer crash logs already document the failure mode.
JB Site IFS migration timing overlaps platform build	High	Possible	IFS adapter stub in Sprint 4 validates interface completeness. Migration can begin from a confirmed contract.
Two-site deployment introduces configuration drift	Medium	Low	Environment files version-controlled in Git (secrets excluded). Deployment runbooks per site in vault.
SM-Portal integration introduces Sprint 1 scope risk	Medium	Possible	Phase 1 Track C surfaces this early. Standalone fallback defined. Sprint 1 is not blocked.
Historical PLMServer data quality too poor to migrate	Medium	Possible	Phase 1 Track A validates data quality. If not viable, archive PLMServer DB read-only.
ISO 13485 audit during build programme	Critical	Low	Software Validation plan starts Phase 2. Behavioural Spec and UAT designed as dual-purpose evidence from Phase 1.

16. Business Case Summary

Benefit	Annual Value	Source
Time savings — BOM processing (13 min → 90 sec)	~\$225,000/year	PLMServer analysis: ~3,000 engineer-hours/year
Better supplier pricing — automated best-price	~\$100,000/year	PLMServer analysis: 2% on \$5M parts spend
Reduced rework from BOM errors	~\$24,000/year	PLMServer analysis
Engineer retention	~\$30,000/year	PLMServer analysis: 2 fewer replacements
ECN efficiency — approval time reduction	Unquantified	Stargile friction eliminated
IFS migration risk reduction	Unquantified — strategic insurance	ERP Adapter pattern
Two-site deployment from one codebase	Unquantified — operational saving	Configuration, not code
Total quantified annual benefit	~\$379,000/year	PLMServer analysis baseline

17. Platform Evolution Roadmap





Each future module adds capabilities to the same platform. No new databases, no new authentication systems, no new event buses. The ERP Adapter established in Iteration 1 serves every module — when JB Site migrates to IFS, all modules migrate simultaneously by changing one environment variable.

Appendix A: Phase Timeline Summary

Phase	Weeks	Gate Deliverables
Phase 1 — Discovery	1–4	ECN Spec + BOM/Supplier Spec + All adapter interfaces + Stargile MI gap analysis + movex-rest-api extension spec + Retired lists

Phase	Weeks	Gate Deliverables
		+ SM-Portal decision + Retention decisions + Two-site model confirmed + Vault entries
Phase 2 — Architecture	5–6	ADR + Shared data model + NFRs + Docker confirmed + Movex sandbox confirmed + Supplier test credentials + movex-rest-api gap endpoints scheduled + Build plan
Phase 3 — Build	7–20	Working platform (3 modules) + IFS adapter stub + JB Site config + Both MAS agents registered + Sprint 3 performance demo + UAT both modules + Training materials
Phase 4 — Cutover	21–24	Go-live (both sites, both modules) + Software Validation (IQ/OQ/PQ — ECN) + Training records + Both rollback procedures rehearsed + Both legacy systems decommissioned + Vault handover

Appendix B: Legacy System Summary

Stargile — ECN (Java)

Attribute	Detail
Location	Internal server — source extraction pending permission
Function	ECN workflow, approval routing, Movex BOM/route commit
ERP integration	Own Java classes calling Movex MI API directly — not movex-rest-api
Users	Production Engineers, Document Control, SMT Engineers, Test Engineers
Compliance	ISO 13485 audit trail and approval records
Key gap	Status 60 exit mechanism unconfirmed — Phase 1 Track A
Key Phase 1 output	MI program inventory + gap analysis vs movex-rest-api

PLMServer — BOM / Supplier Intelligence

(PHP/MySQL/Apache)

Attribute	Detail
Location	c:/Projects/PLM/PLMServer
Function	BOM upload, multi-supplier pricing and availability, BOM approval for Purchasing
ERP integration	Movex via SQL Server view (read-only order data)
Users	Engineers/Planners (2 of 8 active), CBMs, Purchasing
Status	Largely abandoned — NPS -40 — due to 13-minute BOM processing
Security issues	SSL disabled; raw SQL queries; OAuth tokens not refreshed proactively
Analysis assets	analysis/PLMServer_Complete_Analysis.md + analysis/plm-tables.sql

Appendix C: Supplier Integration Reference

Supplier	Auth	Rate Limit	Reliability	Key Notes
DigiKey	OAuth2	1,000 req/day	High	Token refresh failures in PLMServer — Celery proactive refresh in platform
Mouser	API Key	1,000 req/hour	High	Occasional timeouts — retry with exponential backoff
Future Electronics	API Key	Unknown	Medium	Inconsistent response format — adapter normalises
Element14	API Key	100 req/min	Medium	Frequent stock mismatches — staleness warning in UI
Verical / Arrow	API Key	Unknown	Medium	
Octopart	API Key	Unknown	High	Primary source for alternatives + RoHS/REACH compliance data

Appendix D: Architectural Decision Log

All decisions must be recorded in `vault/decisions/` using `/adr` .

Decision	Decision Made	Rationale
Unified platform vs two separate replacements	Unified platform	Shared data model; supplier intelligence needed during ECN; compliance requires joined data; single ERP Adapter for both migrations
Data authority model	Movex SOT for committed BOMs; platform manages change workflow and draft state	Avoids bidirectional sync risk; ISO 13485 auditor expectation; multi-site safe; AI value is in the change history, not the committed state
Two-site deployment	Multi-configuration, not multi-tenancy	Same Scanfil APAC entity; configuration handles ERP adapter per site; country split handled by deployment, not architecture
Stargile ERP interface	MovexRestAdapter calls movex-rest-api (HTTP); Phase 1 Track A produces MI gap analysis for movex-rest-api extension	One ERP boundary for the whole platform; .NET team extends movex-rest-api once; cleaner than Python calling MI directly
Backend language	Python/FastAPI	AI/agent integration native; async architecture solves sequential API bottleneck; deviation from .NET standard justified by AI platform roadmap
Supplier concurrency	Parallel async (asyncio) + Celery worker	asyncio.gather with per-supplier semaphores; 13 min → under 90 sec; Celery for background tasks without blocking API responses
Database	PostgreSQL 16	Replaces MySQL (PLMServer); removes SQL Server dependency; JSONB; ISO 13485 audit support; no licence cost

Decision	Decision Made	Rationale
Deployment	Docker on Windows Server (IT has Docker familiarity)	Reproducible (IQ evidence); environment parity (OQ evidence); rollback by image tag; two-site deployment from same images
Frontend	SM-Portal module (recommended)	Shared auth, consistent UX, platform module pattern. Standalone is the documented fallback pending Phase 1 review.

Appendix E: Project Ecosystem Map

Project	Path	Role
Stargile (ECN — Java)	Internal server — source pending	System being replaced. Phase 1 Track A: MI program inventory + gap analysis.
PLMServer (BOM/Supplier — PHP)	c:/Projects/PLM/PLMServer	System being replaced. Full analysis available.
NEXUS (new)	c:/Projects/Nexus/	Unified platform. Commit hook Sprint 1. Both agents Sprint 5.
movex-rest-api	c:/Projects/MOVEX/API-Integration/movex-rest-api	ERP surface. MovexRestAdapter calls this over HTTP. Extended in Phase 2 to cover Stargile MI gap.
SM-Portal	c:/Projects/SM-Portal	Frontend host candidate. Phase 1

Project	Path	Role
		review confirms integration approach.
Knowledge Vault	c:/Projects/Knowledge-Management	Organisational memory. Used from Phase 1 onward.
WMS	c:/Projects/WMS	Future. Subscriber to bom.revised and ecn.completed events.
MES	Future	Iteration 4. ECN + BOM architecture is the template.